

```
In [4]: import pandas as pd
```

```
In [10]: data = pd.read_csv("/Users/prathikraju/Documents/Scaler case study/Netflix_c
```

```
In [12]: data.head()
```

```
Out[12]:
```

	show_id	type	title	director	cast	country	date_added	release_year
--	---------	------	-------	----------	------	---------	------------	--------------

0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
---	----	-------	-------------------------	-----------------	-----	---------------	--------------------	------

1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
---	----	---------	---------------	-----	---	--------------	--------------------	------

2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
---	----	---------	-----------	-----------------	---	-----	--------------------	------

3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
---	----	---------	-----------------------	-----	-----	-----	--------------------	------

4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
---	----	---------	--------------	-----	---	-------	--------------------	------

```
In [14]: data.shape
print("Shape:", data.shape)
```

```
Shape: (8807, 12)
```

```
In [16]: data.columns
```

```
Out[16]: Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_add  
ed',  
              'release_year', 'rating', 'duration', 'listed_in', 'description'],  
              dtype='object')
```

```
In [18]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 8807 entries, 0 to 8806  
Data columns (total 12 columns):  
#   Column                Non-Null Count  Dtype    
---  -  
0   show_id                8807 non-null   object   
1   type                   8807 non-null   object   
2   title                  8807 non-null   object   
3   director               6173 non-null   object   
4   cast                   7982 non-null   object   
5   country                7976 non-null   object   
6   date_added             8797 non-null   object   
7   release_year           8807 non-null   int64    
8   rating                 8803 non-null   object   
9   duration               8804 non-null   object   
10  listed_in              8807 non-null   object   
11  description             8807 non-null   object   
dtypes: int64(1), object(11)  
memory usage: 825.8+ KB
```

```
In [26]: data.describe(include='all')
```

Out [26]:

	show_id	type	title	director	cast	country	date_added	releas
count	8807	8807	8807	6173	7982	7976	8797	8807.0
unique	8807	2	8807	4528	7692	748	1767	
top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	
freq	1	6131	1	19	19	2818	109	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.0
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.0
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.0
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.0
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.0
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2019.0
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2021.0

In [42]:

data['rating'].unique()

Out[42]:

array(['PG-13', 'TV-MA', 'PG', 'TV-14', 'TV-PG', 'TV-Y', 'TV-Y7', 'R',
 'TV-G', 'G', 'NC-17', '74 min', '84 min', '66 min', 'NR', nan,
 'TV-Y7-FV', 'UR'], dtype=object)

In [48]:

data['rating'].value_counts(dropna = False)

```
Out[48]: rating
TV-MA      3207
TV-14      2160
TV-PG      863
R          799
PG-13      490
TV-Y7      334
TV-Y       307
PG         287
TV-G       220
NR          80
G          41
TV-Y7-FV   6
NaN         4
NC-17       3
UR          3
74 min      1
84 min      1
66 min      1
Name: count, dtype: int64
```

```
In [66]: bad_rows = data['rating'].isin(['74 min','84 min','66 min'])
data[bad_rows][['title', 'type', 'rating', 'duration']]
```

```
Out[66]:
```

	title	type	rating	duration
5541	Louis C.K. 2017	Movie	74 min	NaN
5794	Louis C.K.: Hilarious	Movie	84 min	NaN
5813	Louis C.K.: Live at the Comedy Store	Movie	66 min	NaN

```
In [70]: data.loc[bad_rows, 'duration'] = data.loc[bad_rows, 'rating']
```

```
In [72]: data.loc[bad_rows, 'rating'] = None
```

```
In [74]: data[bad_rows][['title', 'type', 'rating', 'duration']]
```

```
Out[74]:
```

	title	type	rating	duration
5541	Louis C.K. 2017	Movie	None	74 min
5794	Louis C.K.: Hilarious	Movie	None	84 min
5813	Louis C.K.: Live at the Comedy Store	Movie	None	66 min

```
In [78]: data['rating'].isnull().sum()
```

Out[78]: 7

```
In [82]: data['type'] = data['type'].astype('category')
```

```
In [86]: data['type'].dtype
```

Out[86]: CategoricalDtype(categories=['Movie', 'TV Show'], ordered=False, categories_dtype=object)

```
In [90]: data['date_added'] = data['date_added'].str.strip()
```

```
In [94]: data['date_added'] = pd.to_datetime(data['date_added'], errors = 'coerce')
```

```
In [98]: data['date_added'].dtype
```

Out[98]: dtype('<M8[ns]')

```
In [100... data['date_added'].isna().sum()
```

Out[100... 10

```
In [102... data['date_added'].head()
```

Out[102... 0 2021-09-25
1 2021-09-24
2 2021-09-24
3 2021-09-24
4 2021-09-24
Name: date_added, dtype: datetime64[ns]

```
In [110... data['director'].isna().sum() / len(data['director']) * 100
```

Out[110... 29.908027705234474

```
In [108... data['director'].isna().sum()
```

Out[108... 2634

```
In [116... data['type'].value_counts() / len(data['type']) * 100
```

Out[116... type
Movie 69.615079
TV Show 30.384921
Name: count, dtype: float64

```
In [120... data['year'] = data['date_added'].dt.year
```

In [128... `data.groupby(['year', 'type']).size()`

/var/folders/70/fpb7f1jd16lfs40_zf7jzz7r0000gn/T/ipykernel_79690/683730573.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

`data.groupby(['year', 'type']).size()`

Out [128...

year	type	
2008.0	Movie	1
	TV Show	1
2009.0	Movie	2
	TV Show	0
2010.0	Movie	1
	TV Show	0
2011.0	Movie	13
	TV Show	0
2012.0	Movie	3
	TV Show	0
2013.0	Movie	6
	TV Show	5
2014.0	Movie	19
	TV Show	5
2015.0	Movie	56
	TV Show	26
2016.0	Movie	253
	TV Show	176
2017.0	Movie	839
	TV Show	349
2018.0	Movie	1237
	TV Show	412
2019.0	Movie	1424
	TV Show	592
2020.0	Movie	1284
	TV Show	595
2021.0	Movie	993
	TV Show	505

dtype: int64

In [338... `year_type_pct = round(pd.crosstab(data['year'], data['type'], normalize = 'i`
`year_type_pct`

Out [338...

type	Movie	TV Show
year		
2008.0	50.00	50.00
2009.0	100.00	0.00
2010.0	100.00	0.00
2011.0	100.00	0.00
2012.0	100.00	0.00
2013.0	54.55	45.45
2014.0	79.17	20.83
2015.0	68.29	31.71
2016.0	58.97	41.03
2017.0	70.62	29.38
2018.0	75.02	24.98
2019.0	70.63	29.37
2020.0	68.33	31.67
2021.0	66.29	33.71

In [136...

```
data.groupby('year').size()
```

Out [136...

```
year
2008.0      2
2009.0      2
2010.0      1
2011.0     13
2012.0      3
2013.0     11
2014.0     24
2015.0     82
2016.0    429
2017.0   1188
2018.0   1649
2019.0   2016
2020.0   1879
2021.0   1498
dtype: int64
```

In [150...

```
genre_split = data['listed_in'].dropna().str.split(',')
```

```
In [152... all_genre = genre_split.explode().str.strip()
```

```
In [154... all_genre.unique()
```

```
Out[154... array(['Documentaries', 'International TV Shows', 'TV Dramas',  
      'TV Mysteries', 'Crime TV Shows', 'TV Action & Adventure',  
      'Docuseries', 'Reality TV', 'Romantic TV Shows', 'TV Comedies',  
      'TV Horror', 'Children & Family Movies', 'Dramas',  
      'Independent Movies', 'International Movies', 'British TV Shows',  
      'Comedies', 'Spanish-Language TV Shows', 'Thrillers',  
      'Romantic Movies', 'Music & Musicals', 'Horror Movies',  
      'Sci-Fi & Fantasy', 'TV Thrillers', 'Kids' TV',  
      'Action & Adventure', 'TV Sci-Fi & Fantasy', 'Classic Movies',  
      'Anime Features', 'Sports Movies', 'Anime Series',  
      'Korean TV Shows', 'Science & Nature TV', 'Teen TV Shows',  
      'Cult Movies', 'TV Shows', 'Faith & Spirituality', 'LGBTQ Movies',  
      'Stand-Up Comedy', 'Movies', 'Stand-Up Comedy & Talk Shows',  
      'Classic & Cult TV'], dtype=object)
```

```
In [156... len(all_genre.unique())
```

```
Out[156... 42
```

```
In [158... all_genre.value_counts()
```



```
Out[158... listed_in
International Movies      2752
Dramas                    2427
Comedies                  1674
International TV Shows   1351
Documentaries            869
Action & Adventure       859
TV Dramas                763
Independent Movies       756
Children & Family Movies 641
Romantic Movies          616
TV Comedies              581
Thrillers                577
Crime TV Shows           470
Kids' TV                 451
Docuseries               395
Music & Musicals          375
Romantic TV Shows        370
Horror Movies             357
Stand-Up Comedy          343
Reality TV               255
British TV Shows         253
Sci-Fi & Fantasy          243
Sports Movies            219
Anime Series             176
Spanish-Language TV Shows 174
TV Action & Adventure     168
Korean TV Shows          151
Classic Movies           116
LGBTQ Movies             102
TV Mysteries              98
Science & Nature TV       92
TV Sci-Fi & Fantasy        84
TV Horror                 75
Anime Features            71
Cult Movies               71
Teen TV Shows             69
Faith & Spirituality       65
TV Thrillers              57
Movies                   57
Stand-Up Comedy & Talk Shows 56
Classic & Cult TV         28
TV Shows                  16
Name: count, dtype: int64
```

```
In [184... countries_split = data['country'].dropna().str.split(',')
```

```
In [188... all_countries = countries_split.explode().str.strip()
```

```
In [190... all_countries.unique()
```

```
Out[190... array(['United States', 'South Africa', 'India', 'Ghana', 'Burkina Faso',  
      'United Kingdom', 'Germany', 'Ethiopia', 'Czech Republic',  
      'Mexico', 'Turkey', 'Australia', 'France', 'Finland', 'China',  
      'Canada', 'Japan', 'Nigeria', 'Spain', 'Belgium', 'South Korea',  
      'Singapore', 'Italy', 'Romania', 'Argentina', 'Venezuela',  
      'Hong Kong', 'Russia', '', 'Ireland', 'Nepal', 'New Zealand',  
      'Brazil', 'Greece', 'Jordan', 'Colombia', 'Switzerland', 'Israel',  
      'Taiwan', 'Bulgaria', 'Algeria', 'Poland', 'Saudi Arabia',  
      'Thailand', 'Indonesia', 'Egypt', 'Denmark', 'Kuwait',  
      'Netherlands', 'Malaysia', 'Vietnam', 'Hungary', 'Sweden',  
      'Lebanon', 'Syria', 'Philippines', 'Iceland',  
      'United Arab Emirates', 'Norway', 'Qatar', 'Mauritius', 'Austria',  
      'Cameroon', 'Palestine', 'Uruguay', 'Kenya', 'Chile', 'Luxembourg',  
      'Cambodia', 'Bangladesh', 'Portugal', 'Cayman Islands', 'Senegal',  
      'Serbia', 'Malta', 'Namibia', 'Angola', 'Peru', 'Mozambique',  
      'Belarus', 'Zimbabwe', 'Puerto Rico', 'Pakistan', 'Cyprus',  
      'Guatemala', 'Iraq', 'Malawi', 'Paraguay', 'Croatia', 'Iran',  
      'West Germany', 'Albania', 'Georgia', 'Soviet Union', 'Morocco',  
      'Slovakia', 'Ukraine', 'Bermuda', 'Ecuador', 'Armenia', 'Mongolia',  
      'Bahamas', 'Sri Lanka', 'Latvia', 'Liechtenstein', 'Cuba',  
      'Nicaragua', 'Slovenia', 'Dominican Republic', 'Samoa',  
      'Azerbaijan', 'Botswana', 'Vatican City', 'Jamaica', 'Kazakhstan',  
      'Lithuania', 'Afghanistan', 'Somalia', 'Sudan', 'Panama', 'Uganda',  
      'East Germany', 'Montenegro'], dtype=object)
```

```
In [192... len(all_countries.unique())
```

```
Out[192... 123
```

```
In [194... all_countries.value_counts().head(15)
```

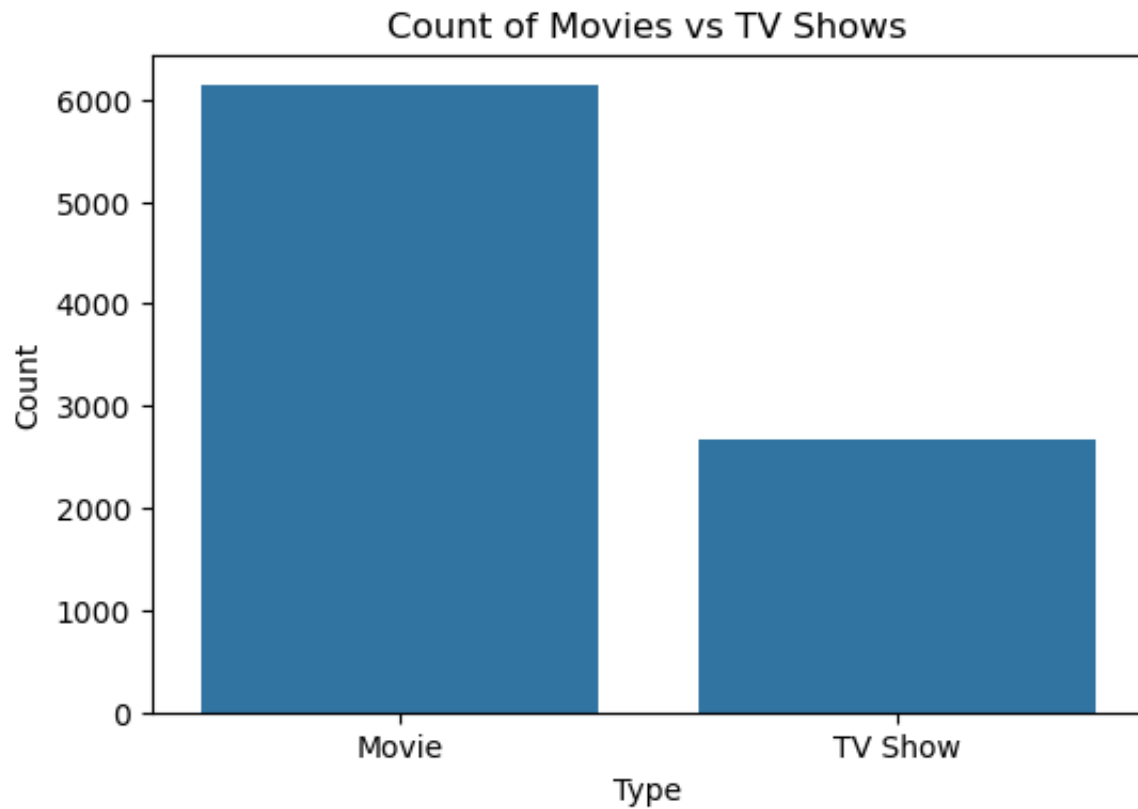
```
Out[194...] country
United States    3690
India            1046
United Kingdom   806
Canada           445
France           393
Japan            318
Spain            232
South Korea      231
Germany          226
Mexico           169
China            162
Australia        160
Egypt            117
Turkey           113
Hong Kong        105
Name: count, dtype: int64
```

```
In [204...] round(data['rating'].value_counts()/ len(data['rating'].dropna()) * 100, 2)
```

```
Out[204...] rating
TV-MA        36.44
TV-14        24.55
TV-PG         9.81
R             9.08
PG-13         5.57
TV-Y7         3.80
TV-Y          3.49
PG            3.26
TV-G          2.50
NR            0.91
G             0.47
TV-Y7-FV      0.07
NC-17         0.03
UR            0.03
Name: count, dtype: float64
```

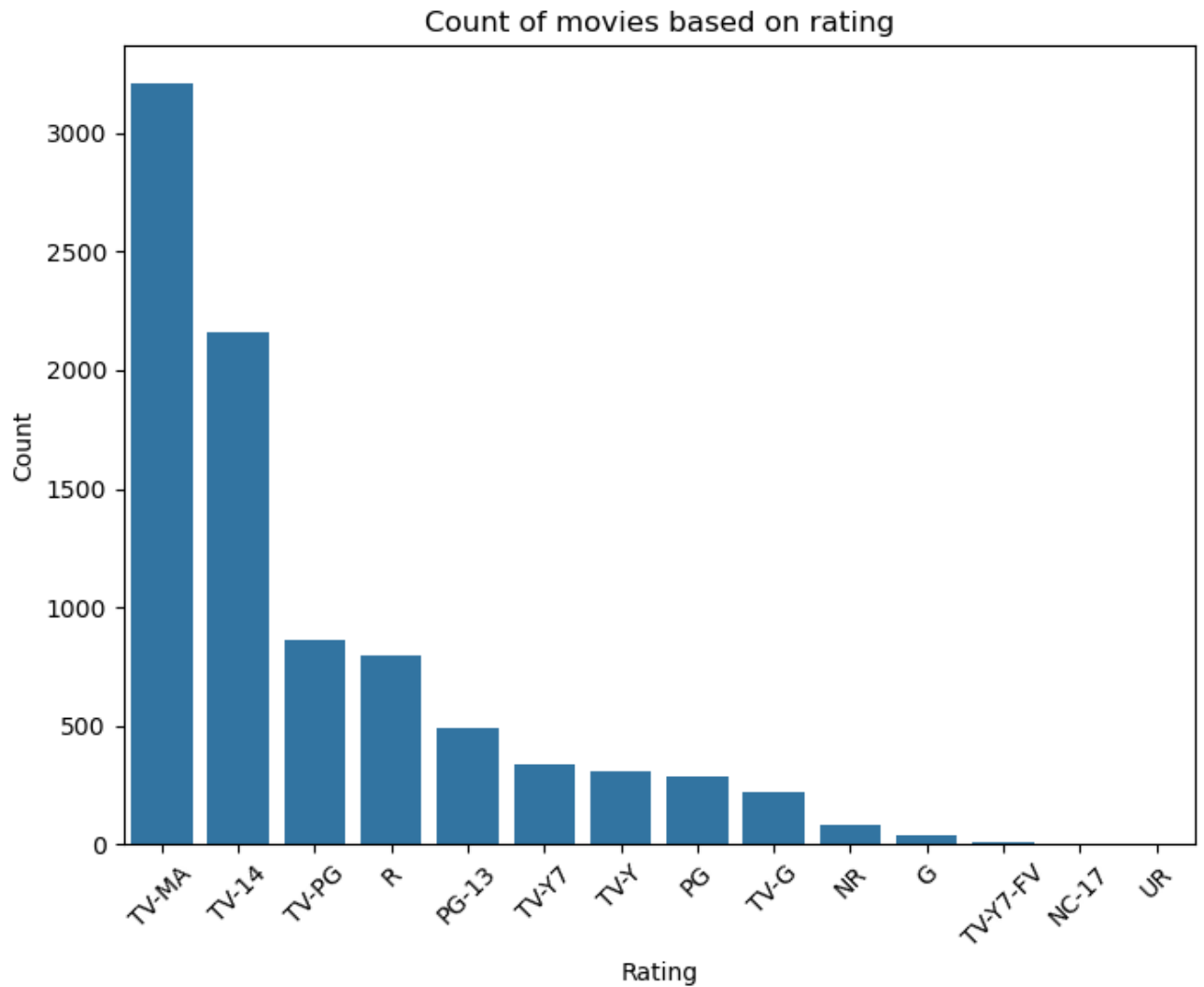
```
In [206...] import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [220...] plt.figure(figsize = (6,4))
sns.countplot(data = data, x = 'type')
plt.title('Count of Movies vs TV Shows')
plt.xlabel('Type')
plt.ylabel('Count')
plt.show()
```



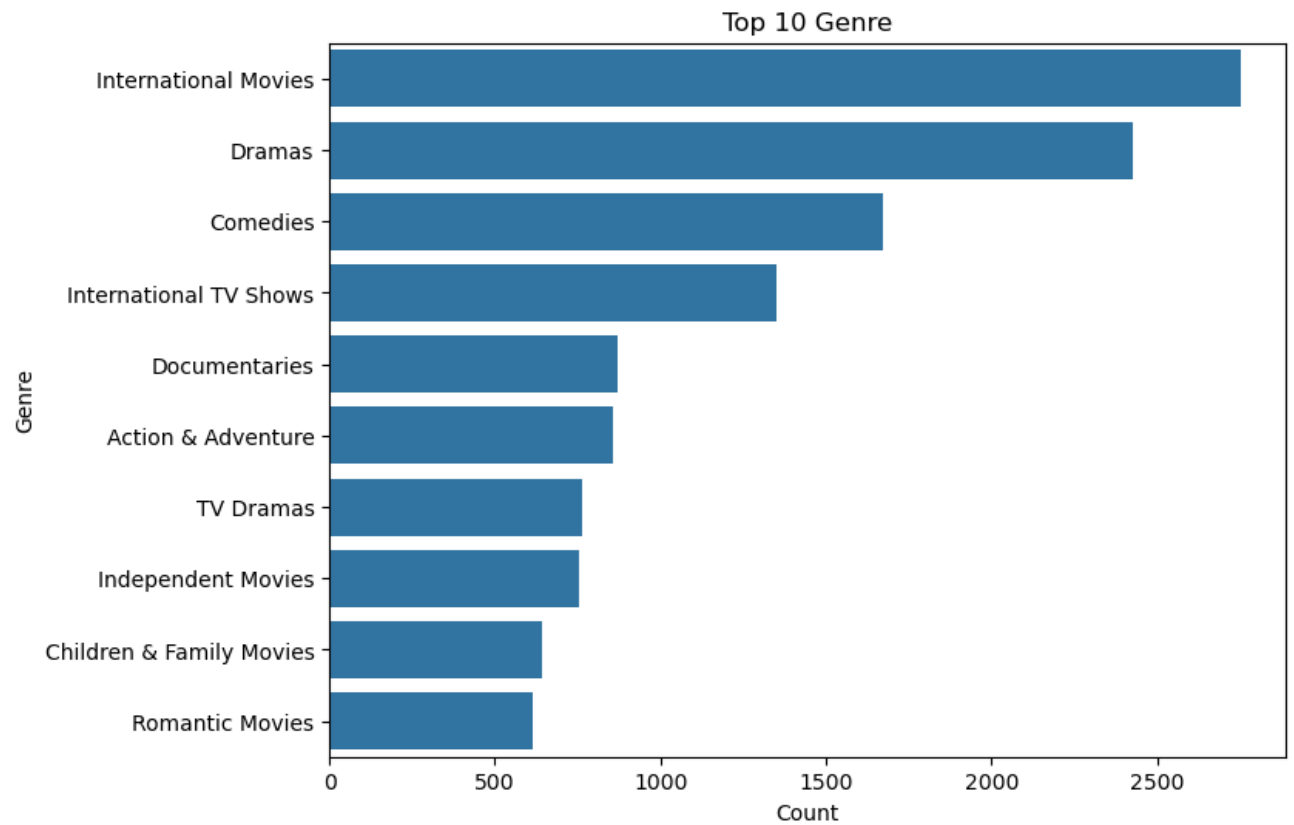
```
In [252... rating_order = data['rating'].value_counts().index

plt.figure(figsize = (8,6))
sns.countplot(data = data, x='rating', order = rating_order)
plt.title('Count of movies based on rating')
plt.xlabel('Rating')
plt.ylabel('Count')
plt.xticks(rotation = 45)
plt.show()
```



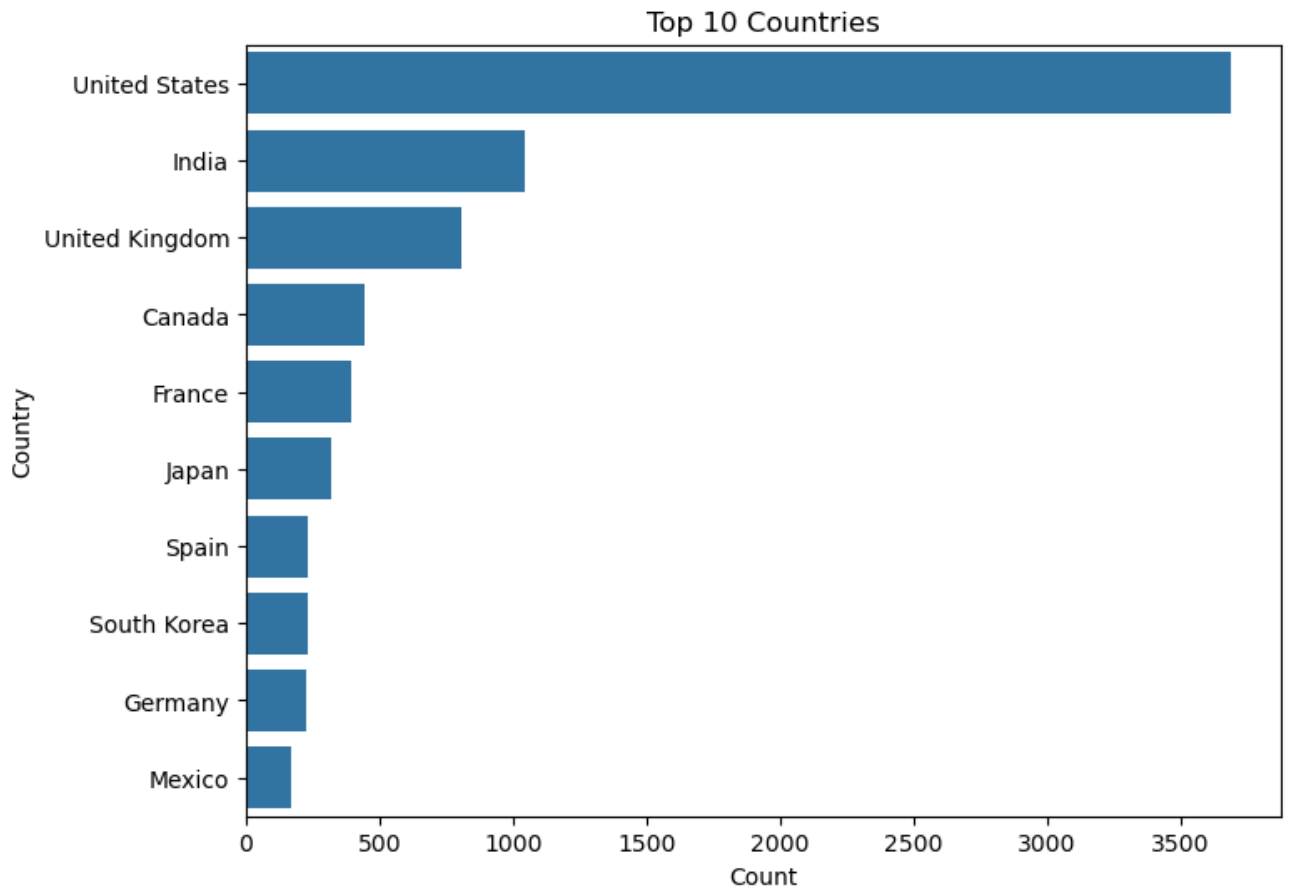
```
In [248... genre_counts = all_genre.value_counts().head(10)

plt.figure(figsize = (8,6))
sns.barplot(x= genre_counts.values, y= genre_counts.index)
plt.title('Top 10 Genre')
plt.xlabel('Count')
plt.ylabel('Genre')
plt.show()
```



```
In [260... country_order = all_countries.value_counts().head(10)

plt.figure(figsize =(8,6))
sns.barplot(x = country_order.values, y = country_order.index)
plt.title('Top 10 Countries')
plt.xlabel('Count')
plt.ylabel('Country')
plt.show()
```



```
In [274...] country_type = data[['country', 'type']].dropna()
```

```
In [276...] country_type['country'] = country_type['country'].str.split(', ')
```

```
In [278...] country_type = country_type.explode('country')
```

```
In [280...] country_type['country'] = country_type['country'].str.strip()
```

```
In [282...] country_type.head()
```

```
Out[282...]
   country  type
0  United States  Movie
1  South Africa  TV Show
4         India  TV Show
7  United States  Movie
7         Ghana  Movie
```

```
In [284...] top10 = ['United States', 'India', 'United Kingdom', 'Canada', 'France',
```

```
'Japan', 'Spain', 'South Korea', 'Germany', 'Mexico']
```

```
In [310... filtered = country_type[country_type['country'].isin(top10)]
country_type_percentage = round(pd.crosstab(filtered['country'], filtered['t
country_type_percentage
```

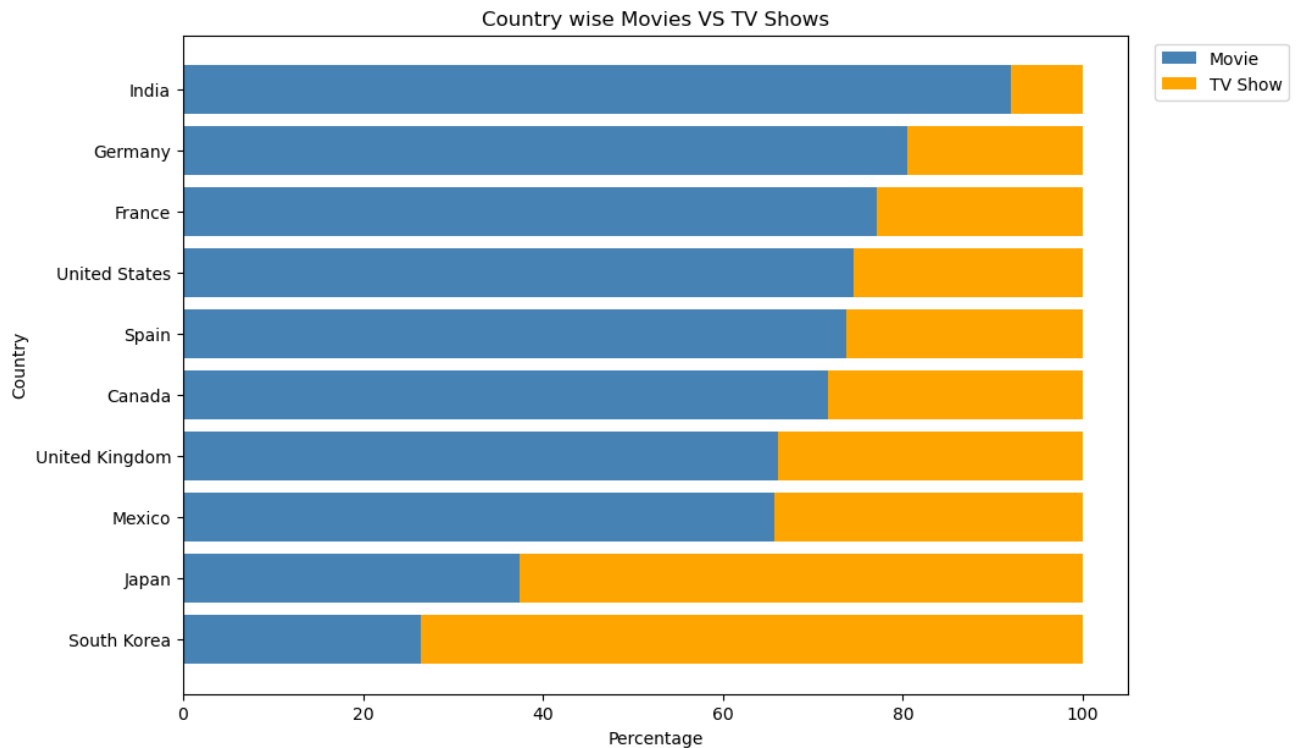
```
Out[310...      type  Movie  TV Show
```

country		
Canada	71.69	28.31
France	77.10	22.90
Germany	80.53	19.47
India	91.97	8.03
Japan	37.42	62.58
Mexico	65.68	34.32
South Korea	26.41	73.59
Spain	73.71	26.29
United Kingdom	66.17	33.83
United States	74.57	25.43

```
In [ ]:
```

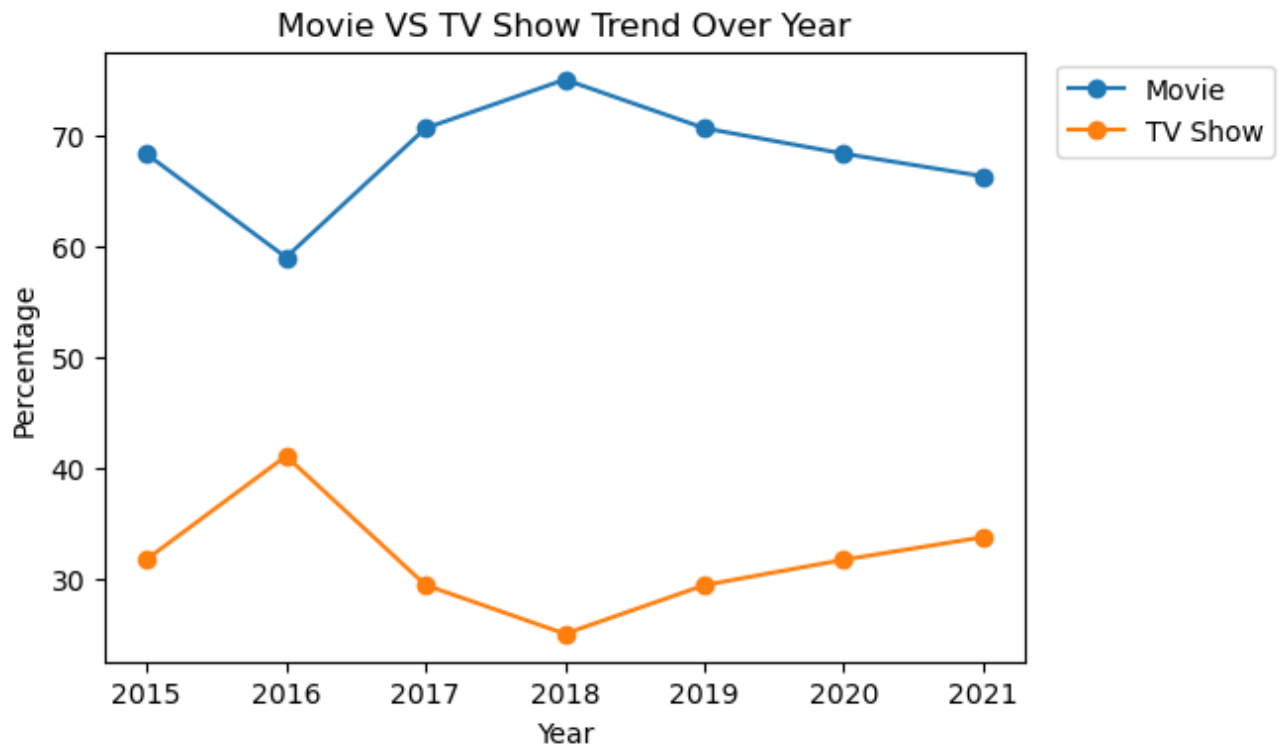
```
In [328... country_type_percentage = country_type_percentage.sort_values('Movie', ascer
```

```
In [332... plt.figure(figsize = (10,7))
plt.barh(country_type_percentage.index, country_type_percentage['Movie'], la
plt.barh(country_type_percentage.index, country_type_percentage['TV Show'],
plt.title("Country wise Movies VS TV Shows")
plt.xlabel('Percentage')
plt.ylabel('Country')
plt.legend(bbox_to_anchor=(1.02, 1), loc='upper left')
plt.show()
```

```
In [350...] year_type_pct_filtered = year_type_pct[year_type_pct.index >= 2015]
```

```
In [354...] plt.figure(figsize = (6,4))
plt.plot(year_type_pct_filtered.index, year_type_pct_filtered['Movie'], mark
plt.plot(year_type_pct_filtered.index, year_type_pct_filtered['TV Show'], ma
plt.title('Movie VS TV Show Trend Over Year')
plt.xlabel('Year')
plt.ylabel('Percentage')
plt.legend(bbox_to_anchor=(1.02, 1), loc='upper left')
plt.show()
```



```
In [358... data['duration_num'] = data['duration'].str.split(' ').str[0].astype(int)
```

```
In [362... data['duration_num'].isnull().sum()
```

```
Out[362... 0
```

```
In [364... data[['type', 'duration', 'duration_num']].head()
```

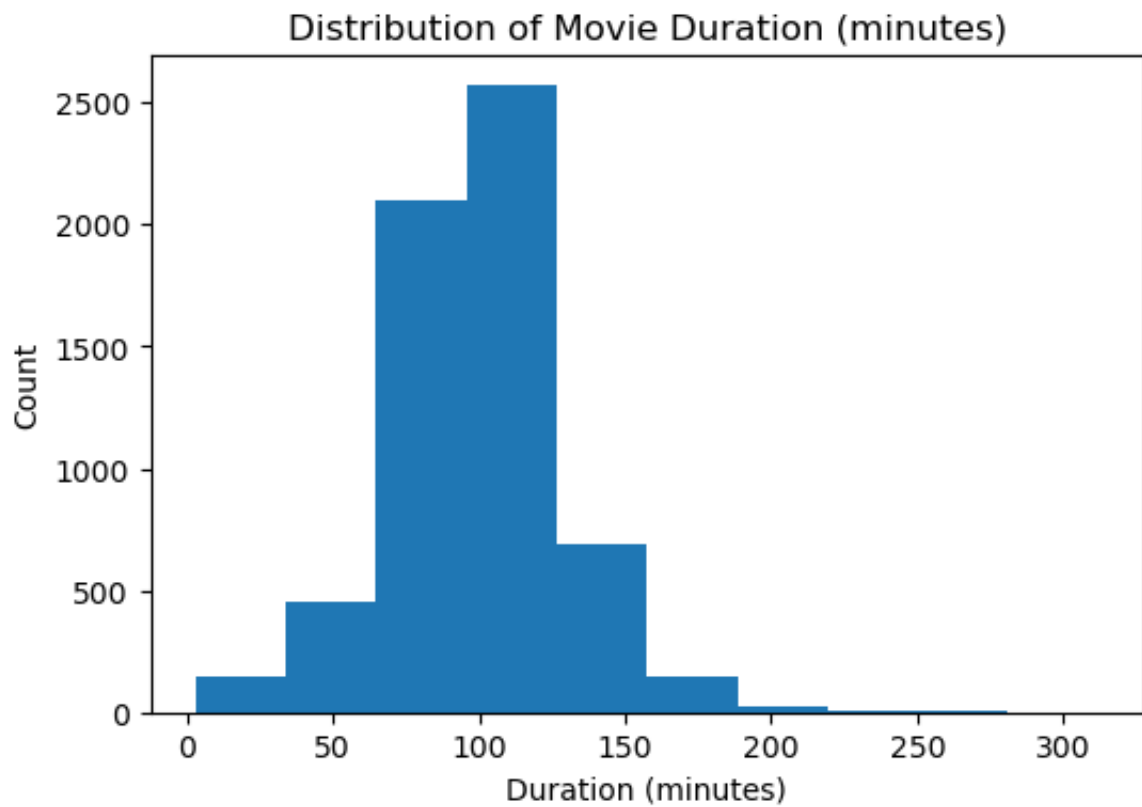
```
Out[364...
```

	type	duration	duration_num
0	Movie	90 min	90
1	TV Show	2 Seasons	2
2	TV Show	1 Season	1
3	TV Show	1 Season	1
4	TV Show	2 Seasons	2

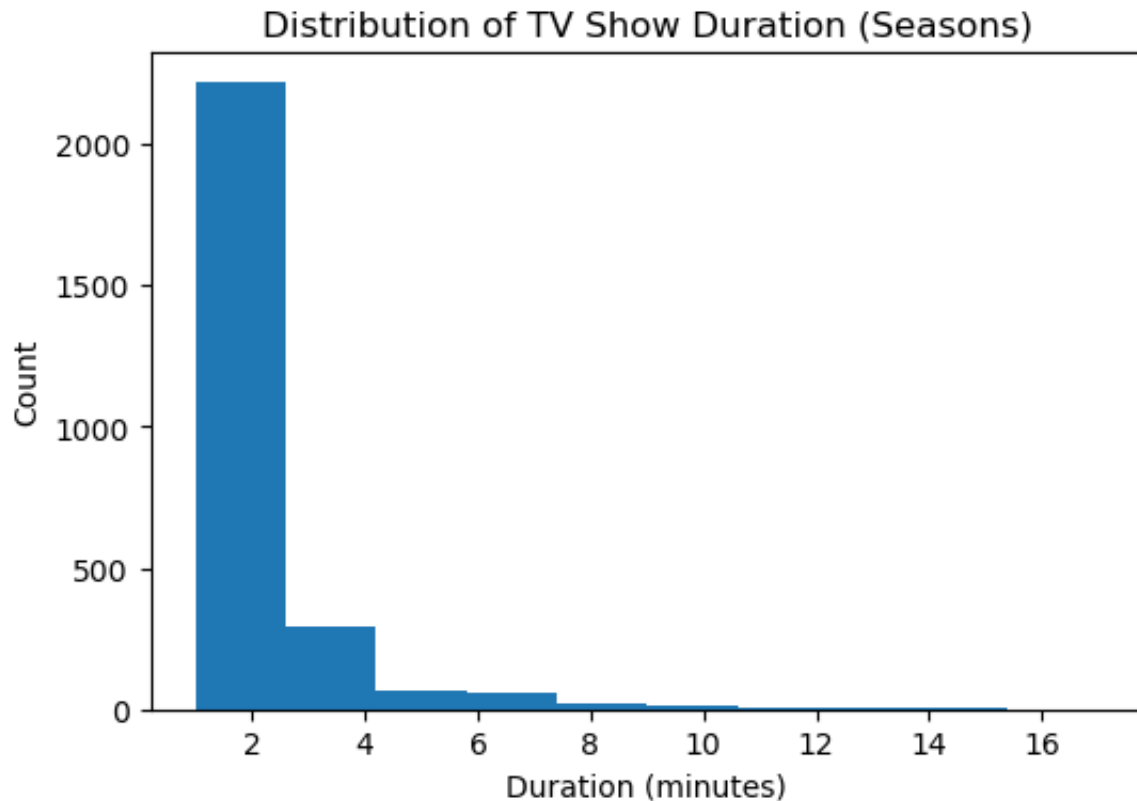
```
In [366... movie_duration = data[data['type'] == 'Movie']['duration_num']
tv_duration = data[data['type'] == 'TV Show']['duration_num']
```

```
In [374... plt.figure(figsize = (6,4))
plt.hist(movie_duration)
plt.title('Distribution of Movie Duration (minutes)')
plt.xlabel('Duration (minutes)')
```

```
plt.ylabel('Count')  
plt.show()
```



```
In [378... plt.figure(figsize = (6,4))  
plt.hist(tv_duration)  
plt.title('Distribution of TV Show Duration (Seasons)')  
plt.xlabel('Duration (minutes)')  
plt.ylabel('Count')  
plt.show()
```



```
In [380... Q1 = movie_duration.quantile(0.25)
Q3 = movie_duration.quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Q1:", Q1, "Q3:", Q3, "IQR:", IQR)
print("Lower bound:", lower_bound, "Upper bound:", upper_bound)

outliers = movie_duration[(movie_duration < lower_bound) | (movie_duration >
print("Number of outliers:", outliers.count())
```

```
Q1: 87.0 Q3: 114.0 IQR: 27.0
Lower bound: 46.5 Upper bound: 154.5
Number of outliers: 450
```

```
In [384... movie_duration.sort_values().head(5)
```

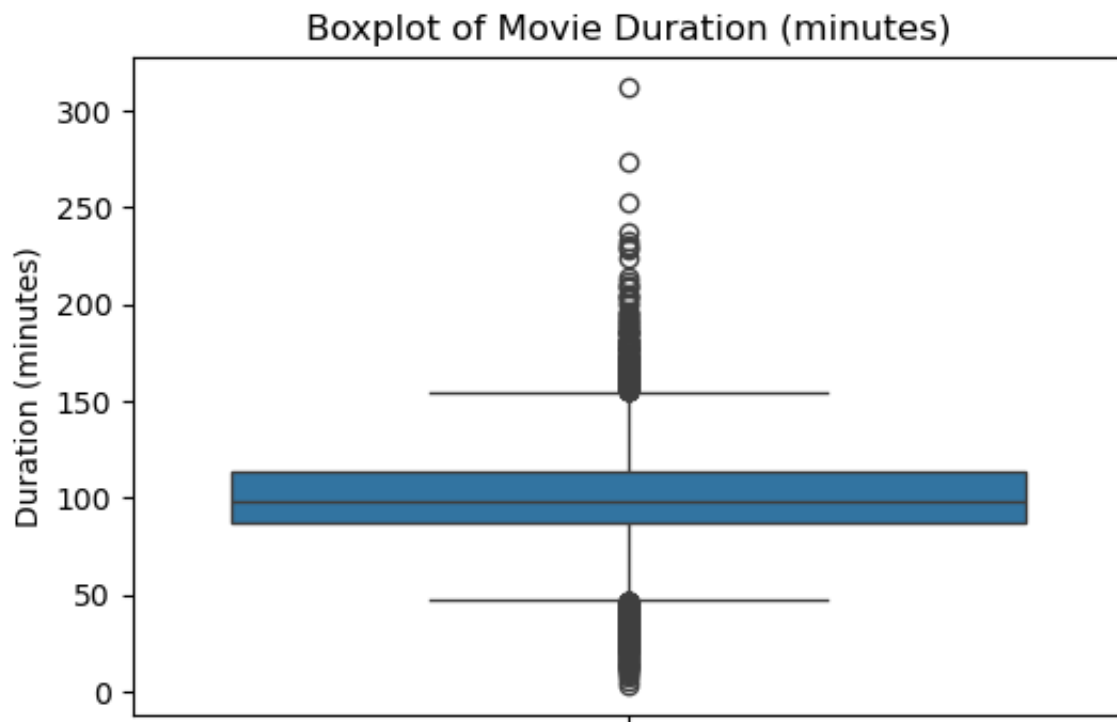
```
Out[384... 3777      3
2713      5
1484      8
1557      9
3535     10
Name: duration_num, dtype: int64
```

In [386... `data.loc[[3777, 2713, 1484, 1557, 3535], ['title', 'type', 'duration', 'list`

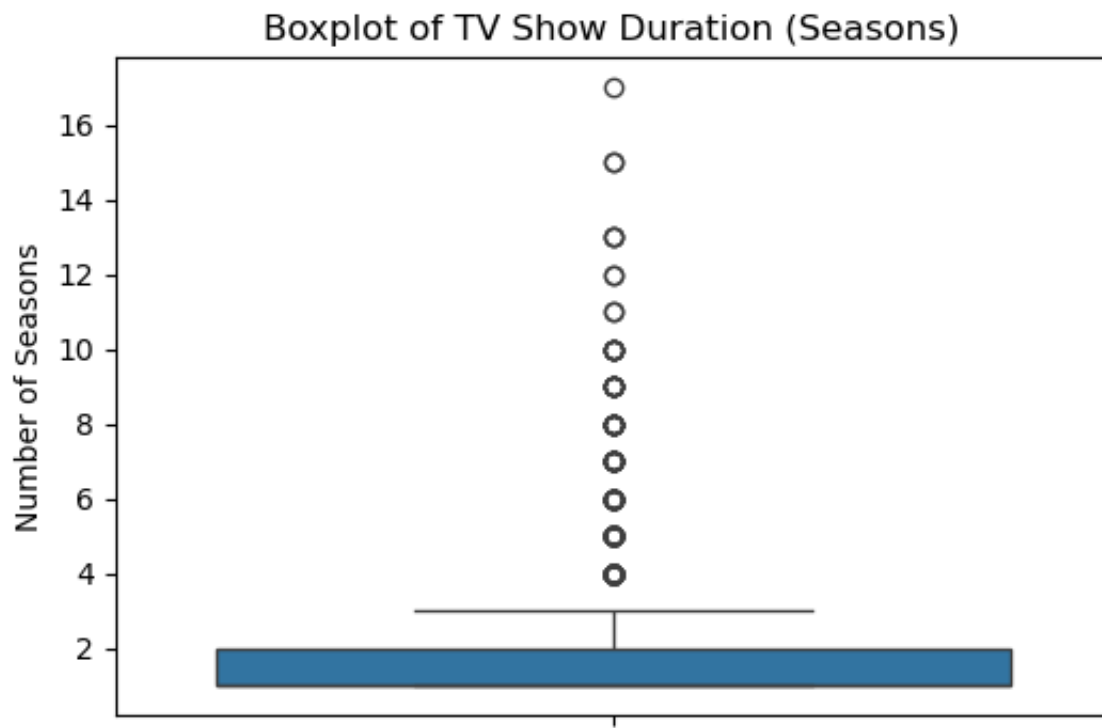
Out [386...

	title	type	duration	listed_in
3777	Silent	Movie	3 min	Children & Family Movies, Sci-Fi & Fantasy
2713	Sol Levante	Movie	5 min	Action & Adventure, Anime Features, Internatio...
1484	Cops and Robbers	Movie	8 min	Dramas
1557	Canvas	Movie	9 min	Children & Family Movies, Dramas
3535	American Factory: A Conversation with the Obamas	Movie	10 min	Documentaries

In [388... `plt.figure(figsize=(6,4))`
`sns.boxplot(y=movie_duration)`
`plt.title('Boxplot of Movie Duration (minutes)')`
`plt.ylabel('Duration (minutes)')`
`plt.show()`



In [390... `plt.figure(figsize=(6,4))`
`sns.boxplot(y=tv_duration)`
`plt.title('Boxplot of TV Show Duration (Seasons)')`
`plt.ylabel('Number of Seasons')`
`plt.show()`



In []:

Netflix Content Strategy Case Study

About the Data

I used the Netflix Movies and TV Shows dataset. It has 8,807 rows and 12 columns. Each row is one title on Netflix, either a movie or a TV show. The columns include show id, type, title, director, cast, country, date added, release year, rating, duration, listed in (genre), and description.

Some columns had missing values. Director was missing in about 30 percent of rows. Cast was missing in about 9 percent of rows. Country was missing in about 9 percent of rows. Date added was missing in only 10 rows. Rating was missing in a handful of rows too.

What I Did (Step by Step)

Step 1. I loaded the data and looked at its basic shape. I used `df.info()` and `df.describe(include='all')` to understand what columns exist, what type of data each one holds, and where anything looked off.

Step 2. I checked data quality. I found that three rows had a strange problem. The rating column had values like 74 min instead of an actual rating like PG or TV MA. I figured out these values had shifted over from the duration column by mistake. I fixed this by moving the value back to duration and marking rating as truly missing for those rows.

I also converted the date added column from plain text into a real date format so I could pull out the year and month later.

Step 3. I decided what to do about missing values. Instead of deleting rows just because one column was empty, I thought about which columns actually mattered for my questions. Since my focus was on content type, genre, country, and rating, I left director and cast missing values alone since they were not part of what I wanted to study. For country, since I did want to study it, I only removed missing values at the exact moment I used that column, not from the whole dataset.

Step 4. I ran non graphical analysis. This means counting things and turning counts into percentages, without charts yet. I did this for type, rating, genre, and country.

Step 5. I unnested columns that held more than one value. Columns like country and genre had entries like United States, India or Dramas, International Movies packed into a single cell. I split these apart and spread them into separate rows so each one could be counted properly on its own.

Step 6. I made charts. I built charts for type, rating, genre, and country separately. Then I built charts that compared two things together, type versus country and type versus year. I also cleaned up the duration column, splitting it into movie minutes and TV show seasons, and built histograms and boxplots for both.

Step 7. I checked for outliers. I used the IQR method on movie duration and found 450 movies fall outside the normal range. I looked closely at the shortest ones and confirmed they are real short films, not data errors, so I kept them in the dataset.

What I Found

Movies and TV Shows. About 70 percent of Netflix titles are movies and 30 percent are TV shows. This has stayed fairly steady over the years, though the more recent years, from 2018 to 2021, show a slow but steady shift where TV shows are slowly gaining ground.

Ratings. Around 70 percent of everything on Netflix is rated for teens or above. TV MA and R rated content alone make up close to half the catalog. Family friendly ratings like G, PG, and TV Y only make up about 10 percent. This tells me Netflix mainly targets grown up and older teen audiences, with family content playing a smaller supporting role.

Genres. The most common genre tags are International Movies, Dramas, Comedies, and International TV Shows. This tells me Netflix leans heavily on broad, widely appealing stories rather than narrow niche genres. Documentaries and Independent Movies also show up in the top ten, which tells me Netflix invests in more serious or artistic content too, not just mainstream entertainment.

Countries. The United States has by far the most titles, followed by India, United Kingdom, Canada, and France. But something interesting shows up when I look closer. Most countries lean heavily toward movies. India for example is 92 percent movies. But South Korea and Japan flip this pattern completely. South Korea is about 74 percent TV shows and Japan is about 63 percent TV shows. This likely reflects how K dramas and anime series are naturally built as ongoing shows rather than single movies.

Trend over time. Looking year by year from 2015 onward, since earlier years had too few titles to trust, movies peaked in 2018 at about 79 percent of that year's additions. Since then, the share of movies has been slowly falling every year, down to about 67 percent by 2021, while TV shows have been slowly climbing.

Movie and TV show duration. Movie duration ranges from as short as 3 minutes to as long as around 300 minutes. Most movies fall between 87 and 114 minutes, which is a normal length for a feature film. TV show duration ranges from 1 season all the way up to 16 seasons, but most shows only have 1 or 2 seasons, showing Netflix mostly hosts newer or shorter run shows rather than long running series.

Insights Based on Non-Graphical and Visual Analysis

Comments on the Range of Attributes

My dataset has 8,807 titles in total. Release year ranges from 1925 all the way to 2021, showing Netflix carries both very old classic titles and brand new releases.

Movie duration ranges from as short as 3 minutes to as long as around 300 minutes. Most movies fall between 87 and 114 minutes, which is a normal length for a feature film.

TV show duration, measured in seasons, ranges from 1 season all the way up to 16 seasons. Most shows only have 1 or 2 seasons.

Rating has 14 different categories, going from very mild ratings like G all the way to mature ratings like TV MA and NC 17.

Country data covers 123 individual countries once I separated the combined entries, though the United States alone makes up a huge share compared to everyone else.

Comments on the Distribution of Variables and Relationship Between Them

Movies make up about 70 percent of the catalog and TV shows make up about 30 percent. This split has stayed fairly steady overall, though the more recent years show TV shows slowly gaining ground.

Movie duration follows a fairly normal shape with a long tail on the right side. Most movies sit around 90 to 100 minutes with a small number of much longer movies stretching the distribution out.

TV show duration is very heavily skewed. Almost all shows sit at 1 or 2 seasons, with only a small number of shows going much higher than that. This tells me Netflix mostly hosts newer or shorter run shows rather than long running series.

Rating shows a strong lean toward mature content. TV MA and R rated titles alone make up close to half of everything on the platform. Family friendly ratings only make up around 10 percent, so Netflix mainly serves older teens and adults.

Genre shows that international content, dramas, and comedies are the most common categories. This tells me Netflix leans on broad appeal stories rather than narrow niche genres.

When I look at country and type together, something interesting shows up. Most countries lean heavily toward movies, with India reaching 92 percent movies. But South Korea and Japan flip this pattern completely, with both countries leaning heavily toward TV shows instead. This shows me that content strategy is not the same everywhere and it depends a lot on what each country's audience prefers.

When I look at year and type together, movies peaked at about 79 percent of new titles in 2018. Since then, the share of movies has been slowly falling every year while TV shows have been slowly climbing. This looks like a real ongoing shift rather than random yearly noise.

Comments for Each Univariate and Bivariate Plot

Type countplot. This plot shows movies clearly outnumber TV shows, matching my earlier calculation of about 70 percent movies and 30 percent TV shows.

Rating countplot. This plot shows a steep drop from TV MA at the top all the way down to rare ratings like NC 17 and UR. It visually confirms that mature rated content dominates the platform.

Genre barplot. This plot shows International Movies and Dramas as the two largest genres by a wide margin, followed by Comedies and International TV Shows. It shows Netflix invests most heavily in broad, widely liked categories.

Country barplot. This plot shows the United States far ahead of every other country, with India as a clear second place. The drop off after the top few countries is steep, showing content is concentrated in a handful of major markets.

Type versus country stacked bar chart. This plot shows each country's movie and TV show split side by side. It clearly highlights how South Korea and Japan stand out from every other country by leaning toward TV shows instead of movies.

Type versus year line chart. This plot shows the yearly movement of movie and TV show percentages. It clearly shows the rise in 2018 followed by a steady decline in movie share and a steady rise in TV show share through 2021.

Movie duration histogram. This plot shows most movies clustering around 90 to 100 minutes with a long tail stretching toward longer movies. This confirms the normal expected shape for movie runtimes.

TV show duration histogram. This plot shows an extremely tall bar at 1 season with almost nothing after that. This confirms most Netflix shows only run for a single season.

Movie duration boxplot. This plot shows the middle 50 percent of movies packed between about 87 and 114 minutes, with a spread of outlier dots reaching up toward 300 minutes. I checked the shortest of these titles and confirmed they are real short films, not data errors.

TV show duration boxplot. This plot shows an extremely compressed box near 1 to 2 seasons, with a long vertical line of outlier dots reaching up to 16 seasons. This visually proves that long running shows are rare exceptions on Netflix, not the normal case.

Recommendations

Based on everything above, here is what I would suggest to Netflix.

Keep growing TV show investment, especially in markets like South Korea and Japan. Since these markets already lean heavily toward series content, and series tend to keep people subscribed longer through multiple episodes and seasons, doubling down here fits both audience taste and business goals around retention.

Keep investing in international content, but pay attention to which format fits each market. Since different countries have very different natural preferences between movies and TV shows, a one size fits all content strategy would not work well. Content buying and production decisions should be shaped by what actually works in each specific country, not applied the same way everywhere.

Consider growing family friendly content a little more. Since only about 10 percent of the catalog fits family ratings, there may be room to grow this share, especially if Netflix wants to attract more household subscriptions where parents are choosing content for the whole family.

Watch the multi year shift toward TV shows closely. Since this has been a steady trend since 2018 and not just a one time blip, it looks like a real strategic direction rather than random yearly noise. Netflix's content buying team should keep tracking this to decide how much further to lean into series over films going forward.

Keep an eye on very long running TV shows. Since almost all shows only last 1 or 2 seasons, the handful that go on for many seasons may represent Netflix's most reliable, most loved series. Understanding what makes these shows different could help guide which new series get renewed for the long term.